

Graph Theory and Applications

Session 3 — Rumours, economic dynamics and Markov chains

Master 1 APE

This session draws extensively on Thomas J. Sargent and John Stachurski's *Intermediate Quantitative Economics with Python* (QuantEcon), particularly the lecture *Finite Markov Chains*. These examples are developed through guided problems and proofs.

1 Problem 1 — A rumour passed from person to person

A message has two possible versions, labelled 0 and 1. It is passed from one person to the next. At each transmission, the version received is kept with probability p , and changed to the other version with probability $1 - p$. These probabilities do not depend on earlier transmissions. Initially the version is 0, and $0 < p < 1$.

After many transmissions, can the last receiver still infer the original version? Let $\psi_t = (\psi_t(0), \psi_t(1))$ be the row vector of probabilities after t transmissions, so $\psi_0 = (1, 0)$.

- 1.1. Draw the directed graph of the two versions, with each arc labelled by its transition probability. Include the self-loops. Write the transition matrix T , in the order $(0, 1)$.
- 1.2. Condition on the present version to prove $\psi_{t+1} = \psi_t T$. Compute ψ_1 and ψ_2 . Why does every row of T sum to one?

Transition matrix and distribution

On a finite state set $S = \{1, \dots, n\}$, a **transition matrix** $P = (P_{ij})$ is **row-stochastic**: $P_{ij} \geq 0$ and $\sum_j P_{ij} = 1$ for every i . The entry P_{ij} is the probability of moving from i to j in one step. A **distribution** is a row vector with nonnegative coordinates summing to one. A **Markov chain** evolves according to these probabilities: conditional on the present state, its next-state distribution does not depend on earlier states. Its distributions satisfy $\psi_{t+1} = \psi_t P$.

- 1.3. For any row-stochastic P , prove that ψP is a distribution whenever ψ is. Prove by induction that $\psi_t = \psi_0 P^t$, including $t = 0$.
- 1.4. Calculate $\det(T - \lambda I)$. Find the eigenvalues and one nonzero column eigenvector for each. Construct an invertible matrix B and a diagonal matrix D such that $T = B D B^{-1}$.
- 1.5. Calculate B^{-1} . Prove by induction that $T^t = B D^t B^{-1}$, then multiply to obtain

$$T^t = \frac{1}{2} \begin{pmatrix} 1 + (2p - 1)^t & 1 - (2p - 1)^t \\ 1 - (2p - 1)^t & 1 + (2p - 1)^t \end{pmatrix}.$$

- 1.6. Deduce the probabilities of the two versions after t transmissions and their limits. Distinguish $p < 1/2$, $p = 1/2$, and $p > 1/2$.
- 1.7. Find a distribution π satisfying $\pi T = \pi$. What happens when $\psi_0 = \pi$? If ψ_t converges for a general transition matrix P , prove that its limit must satisfy the same stationary equation.
- 1.8. Now allow $p = 0$ or $p = 1$. In each case, find all stationary distributions and the trajectory from $(1, 0)$. Does stationarity always describe the limit from every initial distribution?

Stationary distribution

A distribution π is **stationary** for P if $\pi P = \pi$. Starting with this distribution leaves it unchanged at every time. It is a normalised nonnegative *left* eigenvector of P for the eigenvalue 1.

2 Problem 2 — Employment and economic regimes

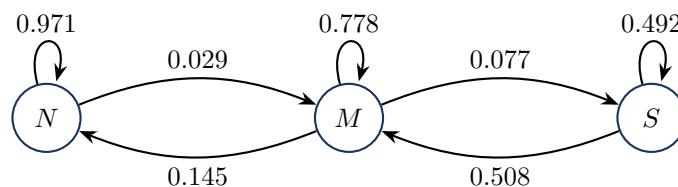
Part I — A model of unemployment

A worker is either employed (E) or unemployed (U). Each month an unemployed worker finds a job with probability α , and an employed worker loses the job with probability β . Assume $0 < \alpha, \beta < 1$, and use the order (E, U) .

- 2.1. Draw the transition graph and write P . Let u_t be the probability of unemployment at month t . Derive an equation for u_{t+1} .
- 2.2. Solve $\pi P = \pi$ and $\pi_E + \pi_U = 1$. Show that at stationarity the flows into and out of unemployment are equal. How does π_U vary with α and β ?
- 2.3. Prove $u_{t+1} - \pi_U = (1 - \alpha - \beta)(u_t - \pi_U)$. Solve the recurrence and prove convergence. Relate the deviation multiplier to the one in the rumour model.

Part II — Hamilton's three economic regimes

Use the monthly model from the notes and QuantEcon. Its states are N (normal growth), M (mild recession) and S (severe recession), with the transition probabilities below.



- 2.4. Write the matrix H in the order (N, M, S) . Verify its row sums. What do the diagonal entries tell us? Can severe recession occur immediately after normal growth?
- 2.5. Starting in normal growth, compute the distributions after one and two months. Why can severe recession have positive probability after two months even though $H_{NS} = 0$?
- 2.6. Find a stationary distribution (x, y, z) . Use the first and third equations to obtain $0.029x = 0.145y$ and $0.077y = 0.508z$, then normalise. Check the remaining equation. What further argument is needed to conclude convergence?

3 Problem 3 — What determines the long-run behaviour?

The rumour at $p = 0$ has a unique stationary distribution but oscillates. Hamilton's model has routes between every pair of states and positive self-loops. We now identify the graph properties behind this difference.

Access and communication

The **transition graph** of P has an arc $i \rightarrow j$ exactly when $P_{ij} > 0$; self-loops are allowed. A **directed walk** follows arcs in their indicated direction. State j is **accessible** from i if such a walk exists, including length zero for $i = j$. States i, j **communicate** if each is accessible from the other. A **communicating class** is a maximal set of states that all communicate. The chain is **irreducible** if every pair of states communicates.

- 3.1. Prove that $(P^t)_{ij} > 0$ if and only if a directed walk of length t leads from i to j . Interpret the entry as a probability, including $t = 0$.

- 3.2.** Prove that communication is an equivalence relation. Identify the classes of the rumour at $p = 0$ and $p = 1$. Is Hamilton's chain irreducible despite its zero entries?

Period and aperiodicity

For a state i with a return walk, put $D(i) = \{t \geq 1 : (P^t)_{ii} > 0\}$. Its **period** is the greatest common divisor of the integers in $D(i)$. A chain is **aperiodic** if every state has return walks and period one. We leave the period undefined for a state with no return walk; this case does not arise in a finite irreducible chain.

- 3.3.** Compute $D(i)$ and the periods in a deterministic two-cycle $a \rightarrow b \rightarrow a$, then a deterministic three-cycle $a \rightarrow b \rightarrow c \rightarrow a$. All arc probabilities are one. Find the stationary distributions. What happens from state a ?
- 3.4.** Why does a positive self-loop imply period one? Apply this to the rumour for $0 < p < 1$, employment, and Hamilton's model. Does one positive self-loop imply irreducibility?
- 3.5.** Let $\Delta^{n-1} = \{x \in \mathbb{R}^n : x_i \geq 0, \sum_i x_i = 1\}$. Use the general simplex version of Brouwer proved in Session 2 to show that every finite transition matrix has a stationary distribution. Verify continuity and preservation of the simplex.
- 3.6.** Distinguish existence, uniqueness and convergence in our examples. Give counterexamples to the claims that existence implies uniqueness, and that uniqueness implies convergence.

The finite-chain theorem — proof in Homework C

Every finite transition matrix has a stationary distribution. Under irreducibility, it is unique. Under irreducibility and aperiodicity, $\psi_0 P^t \rightarrow \pi$ for every initial distribution ψ_0 . Aperiodicity is needed for this convergence conclusion, not for uniqueness under irreducibility.

4 Problem 4 — How can links rank web pages?

Consider three web pages A, B, C . Page A links to B , page B links to C , and C has no outgoing link. A reader at a page with outgoing links chooses one uniformly at random. We seek a score that records where the reader spends time in the long run.

- 4.1.** Draw the directed graph. Why does the uniform-outgoing-link rule by itself fail to define a transition matrix on these three pages? Call a page with no outgoing link a **dangling page**.
- 4.2.** At a dangling page, let the reader choose uniformly among A, B, C . Write the resulting row-stochastic matrix Q , in the order (A, B, C) . Starting at A , where is the reader after one and two steps? Find a stationary distribution of Q . Does it assign zero mass to any page?
- 4.3.** To prevent the reader from being trapped in a small part of the web, at each step choose a link according to Q with probability α , and with probability $1 - \alpha$ jump to page j with probability v_j , where $0 < \alpha < 1$, $v_j > 0$, and $\sum_j v_j = 1$. Write the new transition matrix G . Verify that it is row-stochastic and that every entry is strictly positive.

PageRank in the row-vector convention

For a link matrix Q , a teleportation distribution v and $0 < \alpha < 1$, put

$$G_{ij} = \alpha Q_{ij} + (1 - \alpha)v_j.$$

The **PageRank** vector is a stationary distribution $\pi = \pi G$. We will prove that it is unique and that repeated multiplication by G computes it. A dangling page is first assigned a distribution of destinations so that Q is row-stochastic; in this problem it is $v = (1/3, 1/3, 1/3)$.

4.4. Show that for any distribution x ,

$$xG = \alpha xQ + (1 - \alpha)v.$$

Why is this a self-map of the simplex? Use Brouwer to justify existence of a PageRank vector. Why does Brouwer alone not justify the words *unique* and *computes* above?

4.5. For a row vector $z \in \mathbb{R}^n$, define $\|z\|_1 = \sum_j |z_j|$. Prove that $\|zQ\|_1 \leq \|z\|_1$ whenever Q is row-stochastic. Begin with the triangle inequality in coordinate j , then sum over j .

4.6. For two distributions x, y , show that

$$\|xG - yG\|_1 \leq \alpha \|x - y\|_1.$$

If π and ρ are stationary for G , apply this inequality to prove $\pi = \rho$.

4.7. Set $x_{t+1} = x_t G$. Prove by induction that $\|x_t - \pi\|_1 \leq \alpha^t \|x_0 - \pi\|_1$. Deduce convergence from *every* initial distribution, and explain why every PageRank coordinate is strictly positive.

4.8. If only the ordinary link-following matrix Q had been used, which of your arguments would still work? Give a concrete example showing that uniqueness or convergence can fail for a general Q . Identify the precise role of $\alpha < 1$.

4.9. For the three-page graph, take $v = (1/3, 1/3, 1/3)$ and $\alpha = 1/2$. Compute G , then solve $\pi = \pi G$, $\sum_j \pi_j = 1$. Rank the pages. As a quick check, calculate one iteration from the uniform distribution.

What we have proved

Brouwer gives a stationary distribution for *every* finite transition matrix. With teleportation and $0 < \alpha < 1$, PageRank is the *unique* stationary distribution and the iteration $x_{t+1} = x_t G$ converges to it from any initial distribution.

5 Homework — More examples, Python and a general proof

Homework A and B extend the examples. Homework C is a substantial optional written proof, with a full correction. They are outside the three-hour classroom timing.

Homework A — Social mobility and a four-state chain

Consider L (poor), M (middle class) and R (rich). From L , the next state is always L : this state is **absorbing**. From M , the probabilities of moving to L and R are 0.2 and 0.1; otherwise the state remains M . From R , the probabilities of moving to M and staying R are 0.1 and 0.9.

- A.1.** Draw the graph and write its matrix in the order (L, M, R) . Identify the communicating classes. Which can be left?
- A.2.** Find all stationary distributions. Can a reducible chain have a unique stationary distribution? To determine the limit from the rich state, prove that from either M or R the probability of reaching L within two steps is at least 0.02. Iterate this bound to justify convergence.
- A.3.** Consider arcs $a \rightarrow b, d \rightarrow c$ of probability one, and $b \rightarrow a, b \rightarrow c, c \rightarrow b, c \rightarrow d$ of probability $1/2$. Draw the graph and write its matrix. Exhibit a return to a of length two and exclude odd returns. Deduce its period. Does irreducibility alone guarantee convergence?

Homework B — Return to the examples with Python

Use NumPy for matrix operations. QuantEcon's *Finite Markov Chains* lecture provides further explanations and computational tools.

- B.1.** Compare the rumour's exact formula and repeated matrix multiplication for $p = 0.2, 0.5, 0.8$. Plot $\psi_t(0)$ for $0 \leq t \leq 30$. Repeat for $p = 0$ and $p = 1$.
- B.2.** For Hamilton's matrix, start from $(1, 0, 0)$ and $(0, 0.2, 0.8)$. Plot the three coordinates of the iterated distributions and check the limit from 2.6. Compare with the three-cycle and the social mobility chain. Distinguish a distribution trajectory ψ_t from a simulated sequence of individual states X_t .
- B.3.** Try the following QuantEcon commands on these matrices and compare with your proofs. For employment, retain our order (E, U) ; QuantEcon uses the opposite order.

```
import numpy as np
import quantecon as qe
mc = qe.MarkovChain(P)
print(mc.is_irreducible, mc.is_aperiodic)
print(mc.stationary_distributions)
```

- B.4.** Implement PageRank iteration. For the three-page example, check the limit $(4/17, 6/17, 7/17)$ and compare the errors with $\alpha^t \|x_0 - \pi\|_1$.

Homework C — Prove the finite-chain theorem

Use Brouwer for existence and Question 4.5 for non-expansiveness in $\|\cdot\|_1$.

- C.1.** Assume P irreducible. If a nonzero nonnegative row vector z satisfies $zP = z$, prove that all its coordinates are positive. Choose i with $z_i > 0$ and use $(P^t)_{ij} > 0$ to reach any j .
- C.2.** For stationary distributions π, ρ , put $c = \min_i \pi_i / \rho_i$. Show that $z = \pi - c\rho$ is nonnegative and stationary, with a zero coordinate. Apply C.1 to prove uniqueness. Aperiodicity is not needed.

- C.3.** Now assume aperiodicity as well. Fix a state s . Show that $D(s)$ is nonempty, closed under addition and has gcd one. Select finitely many return lengths $\ell_1, \dots, \ell_r$ with gcd one, justifying finiteness.
- C.4.** By Bézout's identity, for each residue h modulo ℓ_1 , find a nonnegative combination b_h of $\ell_2, \dots, \ell_r$ congruent to h . Replace the integer coefficients by congruent nonnegative ones modulo ℓ_1 . Deduce that every sufficiently large integer belongs to $D(s)$. Treat $\ell_1 = 1$ separately.
- C.5.** Choose walks $i \rightarrow s$ and $s \rightarrow j$. Insert a return walk at s to prove that $A = P^L$ has all entries positive for some integer $L \geq 1$.
- C.6.** Set $\varepsilon = \min_{ij} A_{ij} > 0$ and $\delta = n\varepsilon \leq 1$. If $\delta < 1$, write $A = \delta U + (1 - \delta)R$, where $U_{ij} = 1/n$, and prove that R is row-stochastic. Deduce $\|xA - yA\|_1 \leq (1 - \delta)\|x - y\|_1$ for distributions. Handle $\delta = 1$ directly.
- C.7.** Write $t = kL + r$, $0 \leq r < L$. Use C.6 and non-expansiveness of P^r to prove convergence of xP^t to the unique stationary distribution.